



AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

**Trainer: Aravindraaj.G- Nminds Academy
AWS Academy Certified Educator**

1.Create an EC2 Windows Instance in AWS



Amirthavarshini S
CSE-Artificial Intelligence & Machine Learning,
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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

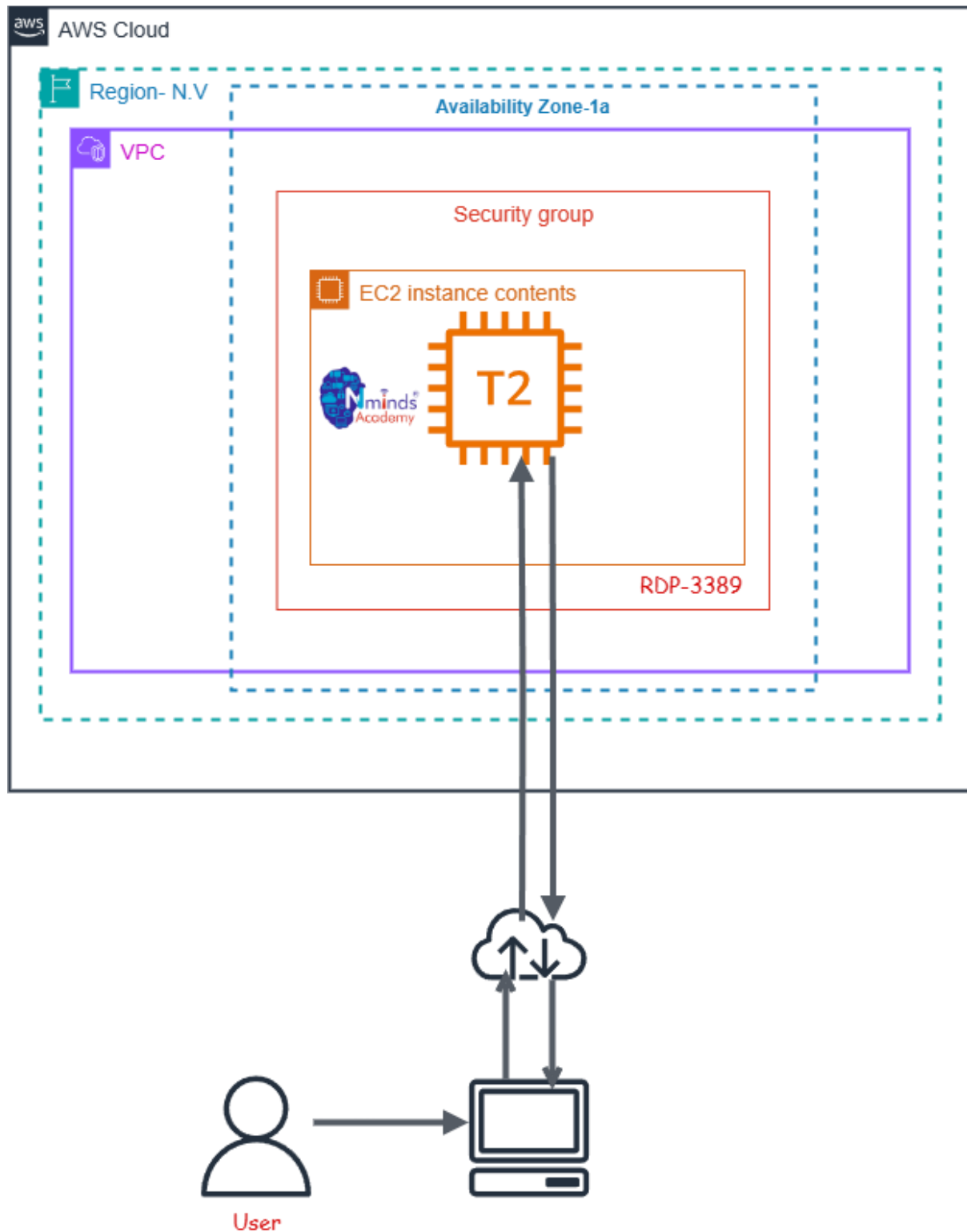
Use Cases for EC2 Windows Instances:

- **Web and Application Hosting:** Run IIS web servers or other Windows-based web applications in the cloud.
- **Database Servers:** Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.
- **Remote Desktop Applications:** Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.
- **Development and Testing:** Create development environments with Windows OS to test .NET or other Windows-based applications.

Used Services in AWS:



Topology



Execution Tasks:

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and the user's name 'Amirtha' with an account ID '3604-3528-3604'. The breadcrumb trail indicates the path: EC2 > Instances > Launch an instance. A green success banner at the top states: 'Success Successfully initiated launch of instance (i-0cbde35c647f2de47)'. Below this, a 'Launch log' section is visible. The 'Next Steps' section provides a search bar and a list of recommended actions: 'Create billing usage alerts', 'Connect to your instance', 'Connect an RDS database', and 'Create EBS snapshot policy'. Each action has a corresponding button and a 'Learn more' link. At the bottom, there are links for 'Manage detailed monitoring', 'Create Load Balancer', 'Create AWS budget', and 'Manage CloudWatch alarms'. The footer contains copyright information and links for 'CloudShell', 'Feedback', 'Console Mobile App', 'Privacy', 'Terms', and 'Cookie preferences'.

The screenshot shows the 'Launch an instance' wizard in the AWS Management Console. The breadcrumb trail is 'EC2 > Instances > Launch an instance'. The main heading is 'Launch an instance' with an 'Info' link. Below the heading, a brief description states: 'Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.' The wizard is divided into two main sections: 'Name and tags' and 'Application and OS Images (Amazon Machine Image)'. The 'Name and tags' section has a text input field with 'Amirtha' and an 'Add additional tags' link. The 'Application and OS Images' section includes a search bar and a 'Quick Start' section with a grid of operating system icons: Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian. The 'Summary' section on the right provides a overview of the configuration: 'Number of instances' is set to 1, 'Software Image (AMI)' is 'Microsoft Windows Server 2025 ...', 'Virtual server type (instance type)' is 't3.micro', 'Firewall (security group)' is 'New security group', and 'Storage (volumes)' is '1 volume(s) - 30 GiB'. At the bottom right, there are 'Cancel' and 'Launch instance' buttons, along with a 'Preview code' link.



aws   Search [Alt+S]  Asia Pacific (Sydney) Amirtha (3604-3528-3604) Amirtha

EC2 > Instances > i-Ocbde35c647f2de47 > Get Windows password

Get Windows password Info

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
 i-Ocbde35c647f2de47 (Amirtha)

Key pair associated with this instance
 amirtha

Private key
 Either upload your private key file or copy and paste its contents into the field below.

[Upload private key file](#)

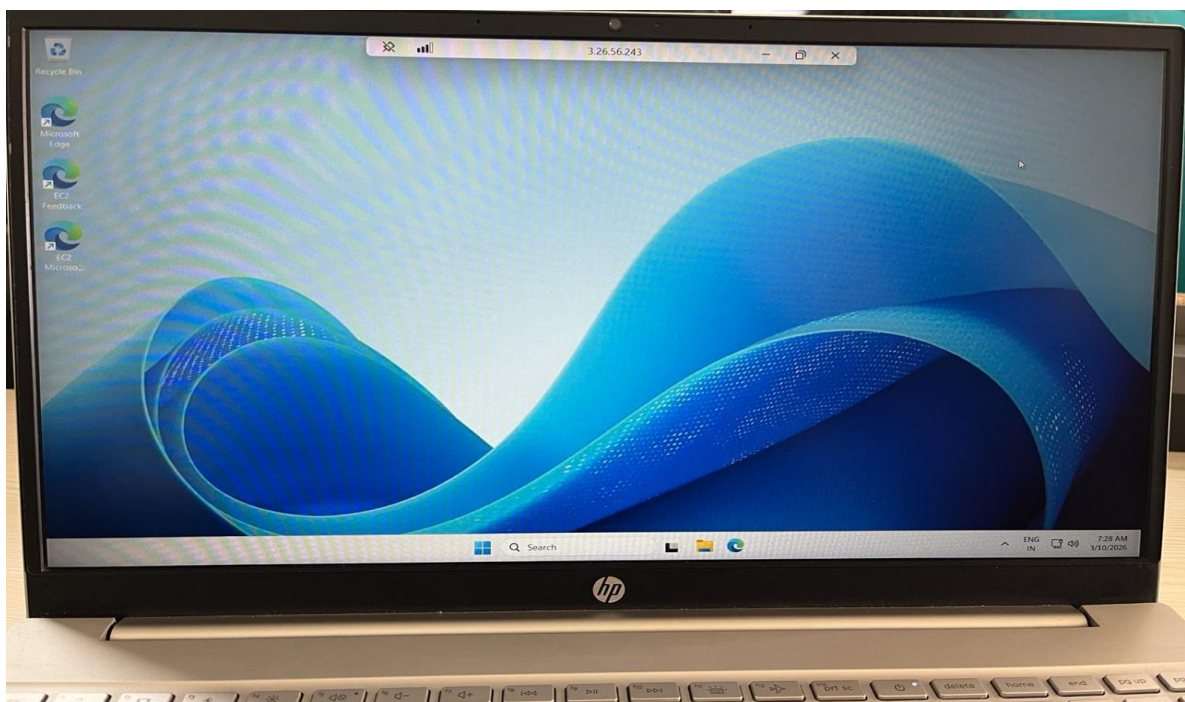
Private key contents

```

HI7oeiroqYtGLq7UlpwKxw+IM/WQIZ1wUcx1bBTmu7P544dCJT0kid1TmiS//QZ
30BQAERDkdRtwk1BW7qsQKBgQCs7gTuHu4HrgVzdQBhA08k/3hVwDrBEZMU1h
RN8vB52bMggcwOhk1sBRvpiA8A69aj6m5L/XIQR65BDb5CDbJnEWqcQ7kx1x54dj
byWijMPxP5rEoKGZUdsqgZu7JcYf6W2F7ek5BTDwZZMvt0yaxNPKI5lorR5+PUN
erol4wKBgGrxOunTJTtKBiQNYVUlrxqDRycm2KtarCGZ4fnZyRwfrpGcl/Rb1moq
L4QJ3pVKPkhdic+XDqGDVNKnhkYwDOFgqvUqF-4y9WZLLSxwdeHUV2qgSuPEQTL
M/knQ5ovGBLK8yZHPgC7nZFkyOAhyzMzDEL4b3oRi/PmvCI7L6cJ
-----END RSA PRIVATE KEY-----

```

[Cancel](#) [Decrypt password](#)





AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Training Capstone Projects

**Trainer: Aravindraaj.G- Nminds Academy AWS
Academy Certified Educator**

2.Configure IIS Webserver on Windows Server in



AWS



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Objective

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

IIS (Internet Information Services) is a web server software created by Microsoft for hosting websites and web applications on Windows Server operating systems. It provides a platform for building, deploying, and managing web-based applications. IIS supports multiple protocols, including HTTP, HTTPS, FTP, and more, and is tightly integrated with the Windows OS.

Common Use Cases of IIS:

1. **Hosting Websites:** IIS is often used by businesses and developers to host websites and web applications, particularly those built with ASP.NET.
2. **Web Applications:** For applications that require robust back-end support, IIS provides features like application pooling and security for web services (e.g., SOAP or RESTful APIs).
3. **Secure File Sharing:** IIS can be configured as an FTP server to securely share files between different locations.
4. **Application Hosting in Enterprises:** Many enterprise applications built on Microsoft technologies rely on IIS for deployment, such as SharePoint, Exchange Server, or Dynamics CRM.
5. **API Hosting:** IIS is commonly used to host APIs (e.g., RESTful services) built with .NET or other technologies.

Used Services in AWS



Used Services in Windows Server



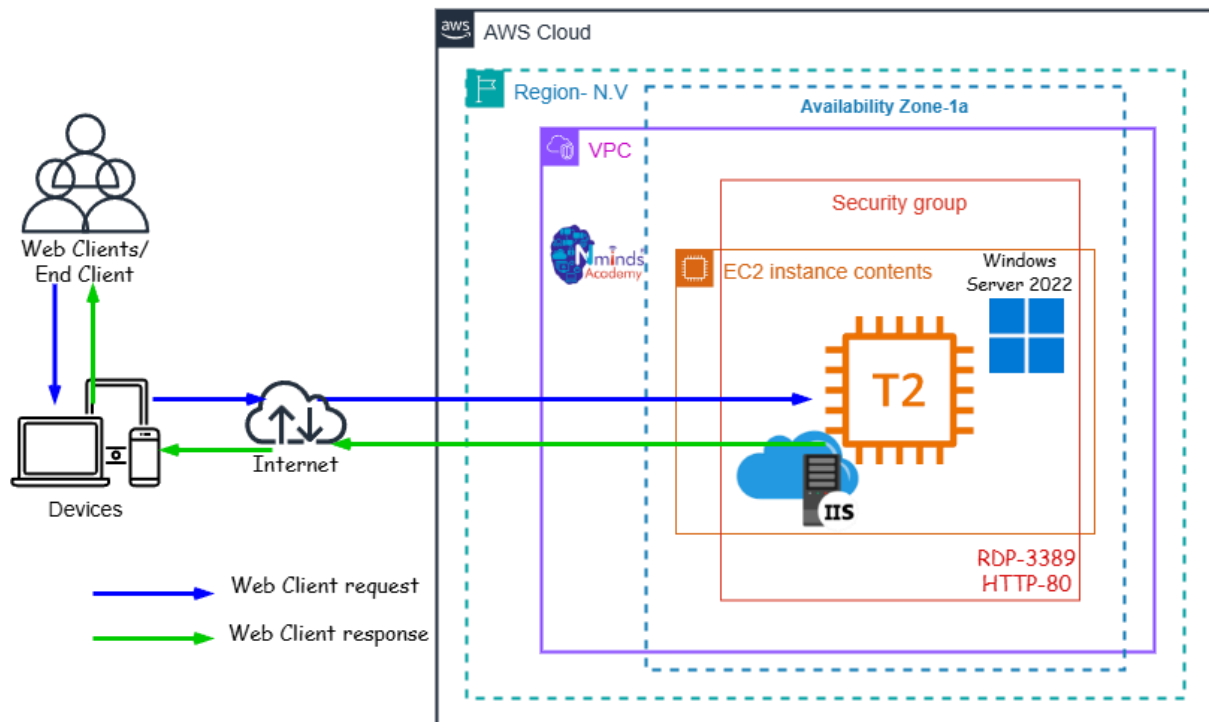
Amazon EC2



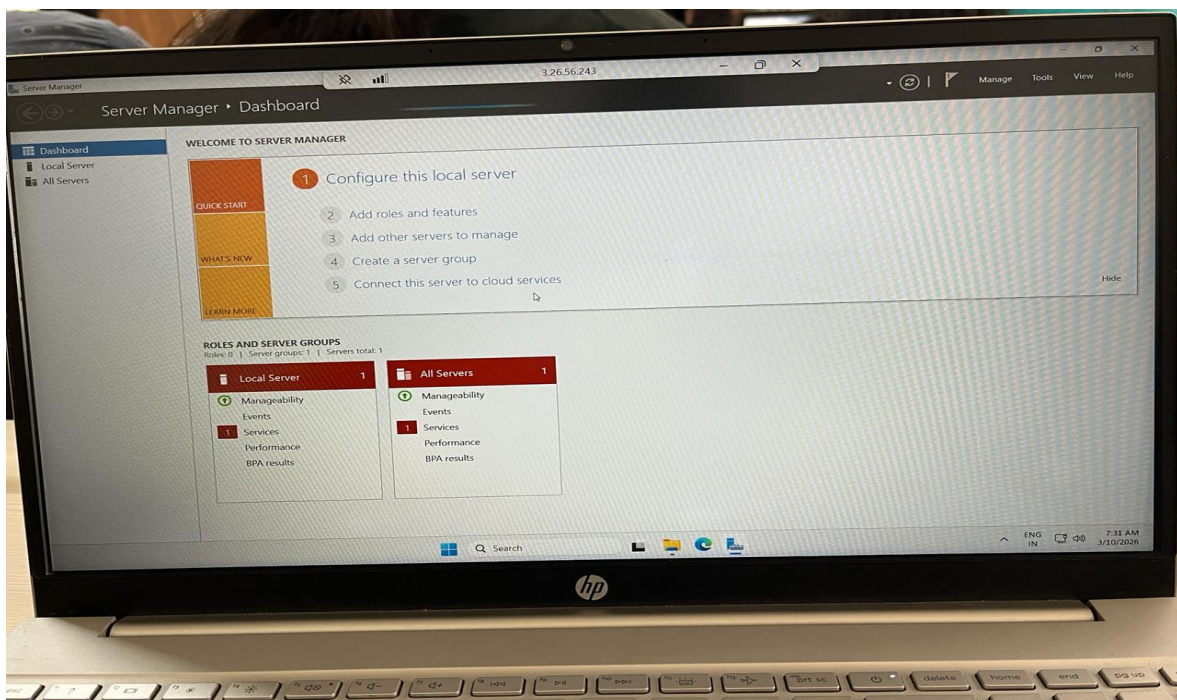
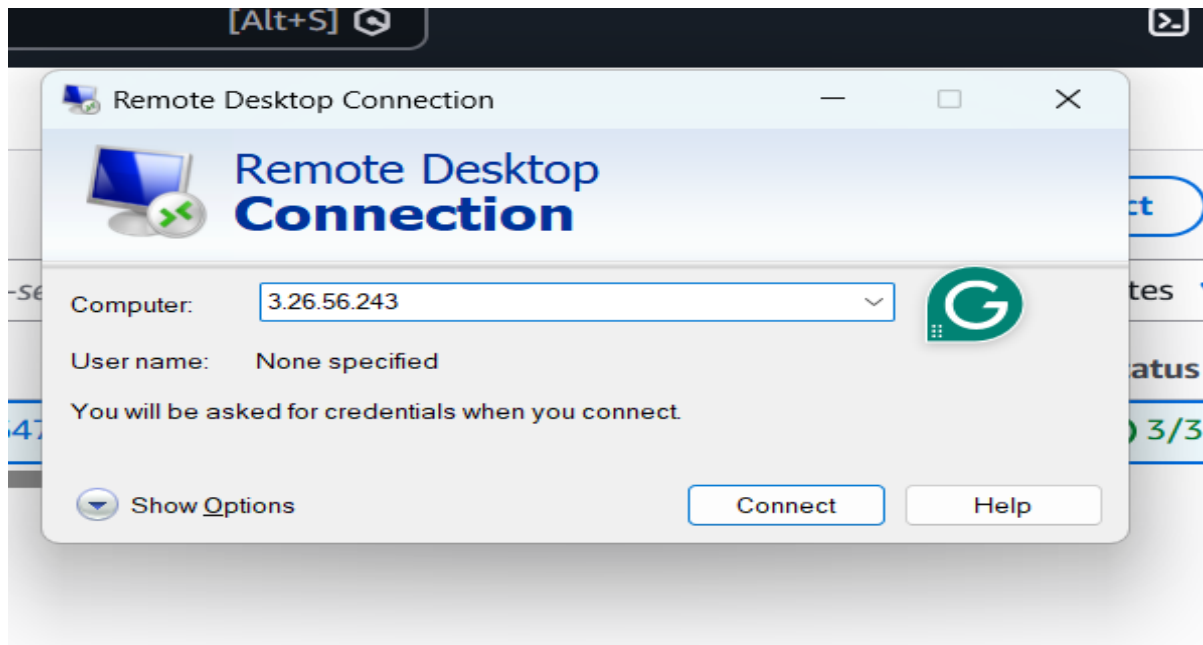
Topology

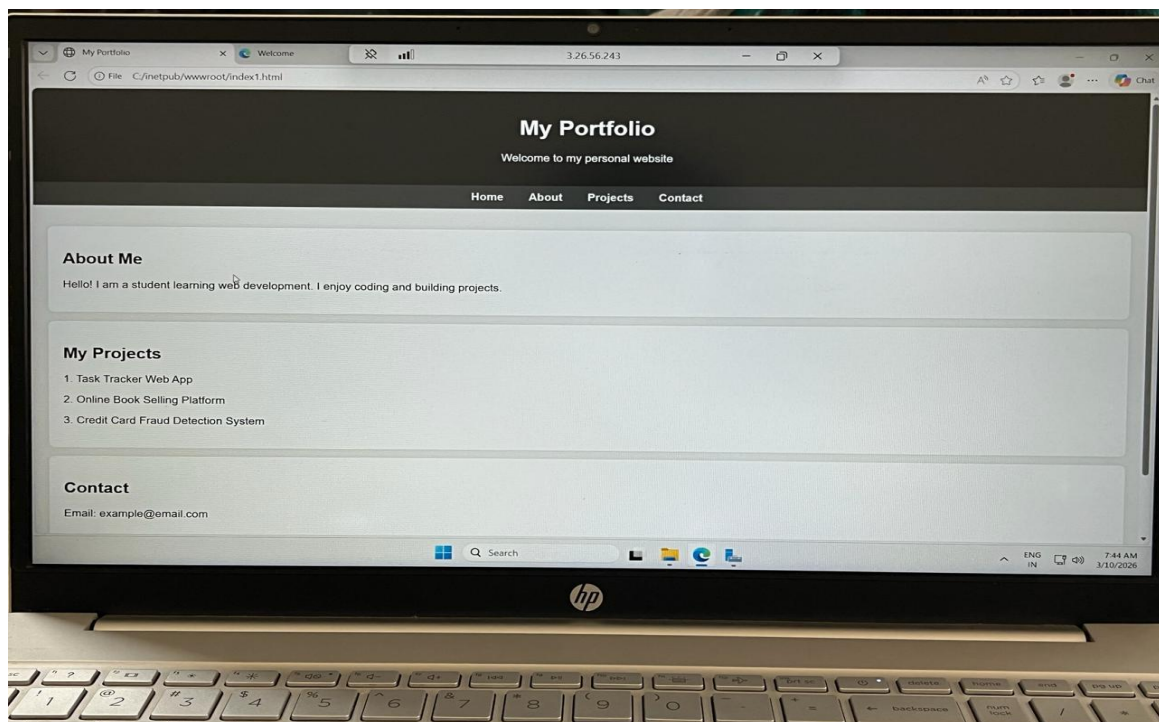
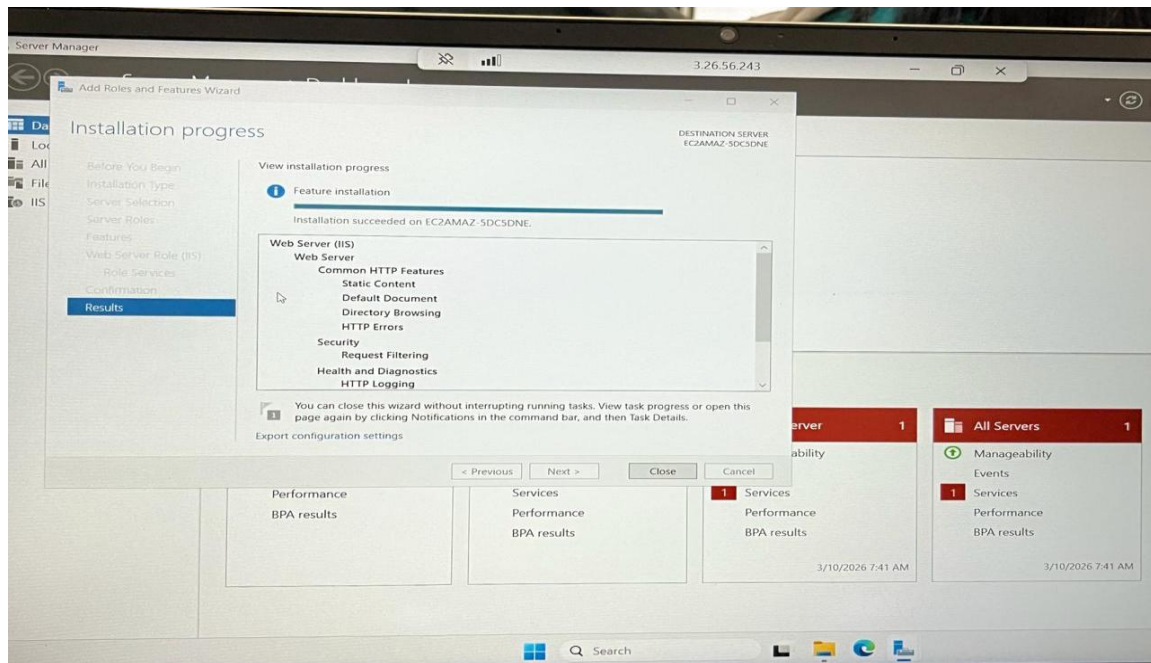
IIS - Internet Information Services

How to Configure the IIS web server in Windows Server with AWS EC2



Execution Tasks:



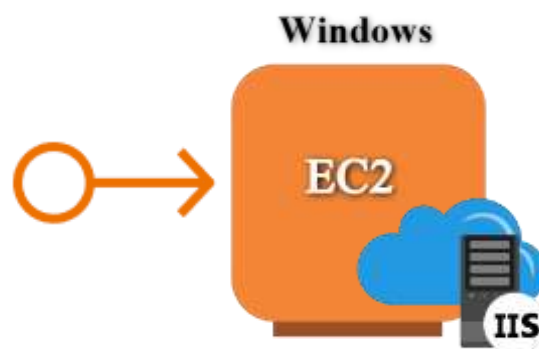




AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

**Trainer: Aravindraj.G- Nminds Academy AWS
Academy Certified Educator**

3.Configure Elastic IP Address to



Windows Web Server in AWS

Amirthavarshini S

**CSE-Artificial Intelligence & Machine Learning,
Sri Eshwar College of Engineering,
Coimbatore.**

Objective



An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

1. Highly Available Applications:

- If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.

2. Web Servers:

- If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.

3. Disaster Recovery:

- Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

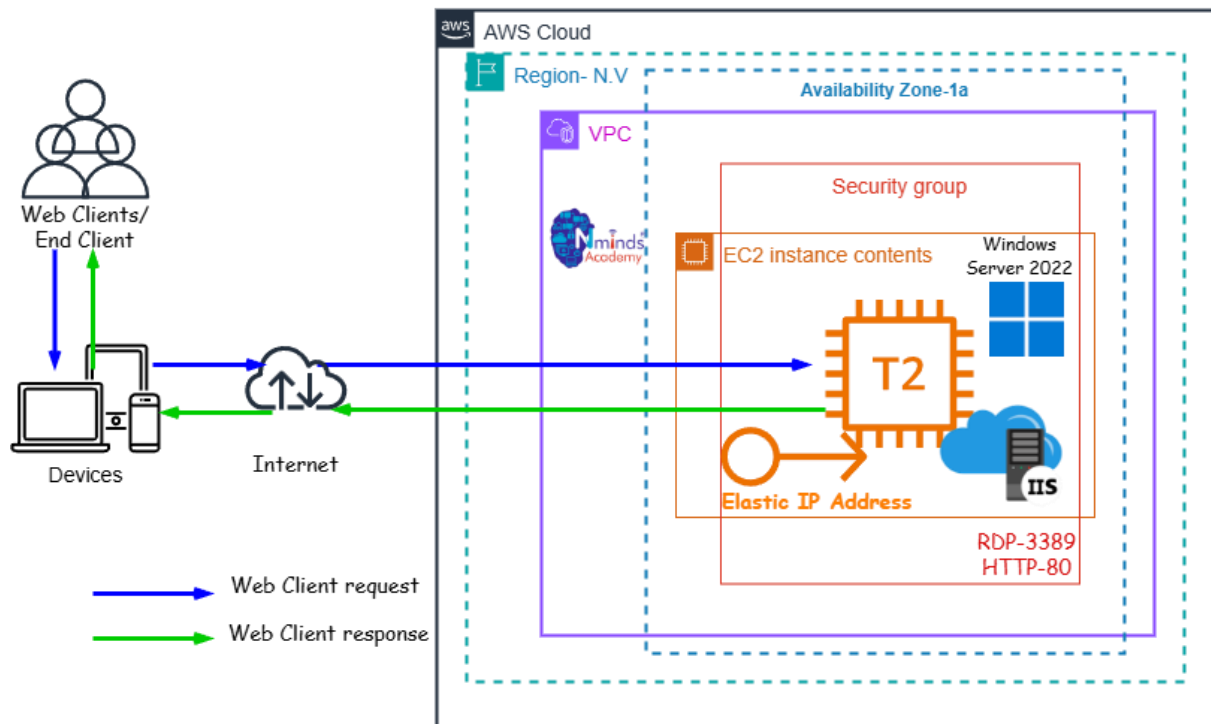
Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

Topology



How to Configure the Elastic IP Address to Windows Web Server with AWS EC2



Used Services in AWS

Amazon EC2

Used Services in Windows Server

IIS - Internet Information Services



Execution Tasks:

ExecutionTasks:



Amazon Elastic IP Address



aws [Search] [Alt+S] Asia Pacific (Sydney) Amirtha (3604-3528-3604) Amirtha

EC2 > Elastic IP addresses > Allocate Elastic IP address

Allocate Elastic IP address [Info](#)

Elastic IP address settings [Info](#)

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
- ☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
- ☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
- ☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [Info](#)

Q ap-southeast-2 X

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tag

aws [Search] [Alt+S] Asia Pacific (Sydney) Amirtha (3604-3528-3604) Amirtha

EC2 > Elastic IP addresses

Elastic IP address associated successfully.

Elastic IP address 52.62.216.80 has been associated with instance i-0cbde35c647f2de47

[Actions](#) [Allocate Elastic IP address](#)

Elastic IP addresses (1) [Info](#)

Find elastic IP addresses by attribute or tag

Public IPv4 address : 52.62.216.80 X Clear filters

☐	Name	Allocated IPv4 addr...	Type	Allocation ID	Reverse DNS record	Assoc
☐	-	52.62.216.80	Public IP	eipalloc-026a16d94edd7aed2	-	i-0cbde35c647f2de47

Select an elastic IP address

[View IP address usage and recommendations to release unused IPs with Public IP insights.](#)



aws

EC2 > Elastic IP addresses

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Settings

Search

[Alt+S]

Asia Pacific (Sydney)

Amirtha (3604-3528-3604)

Amirtha

✓ Elastic IP address allocated successfully.
Elastic IP address 52.62.216.80

Associate this Elastic IP address

Elastic IP addresses (1) Info

Find elastic IP addresses by attribute or tag

Public IPv4 address : 52.62.216.80 Clear filters

☐	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associate
☐	-	52.62.216.80	Public IP	eipalloc-026a16d94edd7aed2	-	-

Select an elastic IP address

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

The image shows a web browser window with a dark-themed header. The browser's address bar at the top shows a back arrow, a forward arrow, a refresh icon, and the text "Not secure 3.27.41.21". On the right side of the header, there are icons for a star (bookmarks) and a download arrow. The main content area has a dark background with the title "My Portfolio" in a large, white, sans-serif font. Below the title, a subtitle reads "Welcome to my personal website". A horizontal navigation bar with a dark background and white text contains four links: "Home", "About", "Projects", and "Contact". The "About" link is currently selected. Below the navigation bar, the page content is divided into three sections, each with a light gray background and a thin dark border. The first section is titled "About Me" and contains the text: "Hello! I am a student learning web development. I enjoy coding and building projects." The second section is titled "My Projects" and contains a numbered list of three items: "1. Task Tracker Web App", "2. Online Book Selling Platform", and "3. Credit Card Fraud Detection System". The third section is titled "Contact" and contains the text: "Email: example@email.com".



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4.Create an EC2 Linux Instance in AWS



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Objective

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An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache, FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- **Web and Application Hosting:** Run Apache web servers or other Linux-based web applications in the cloud.
- **Database Servers:** Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- **Remote Desktop Applications:** Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
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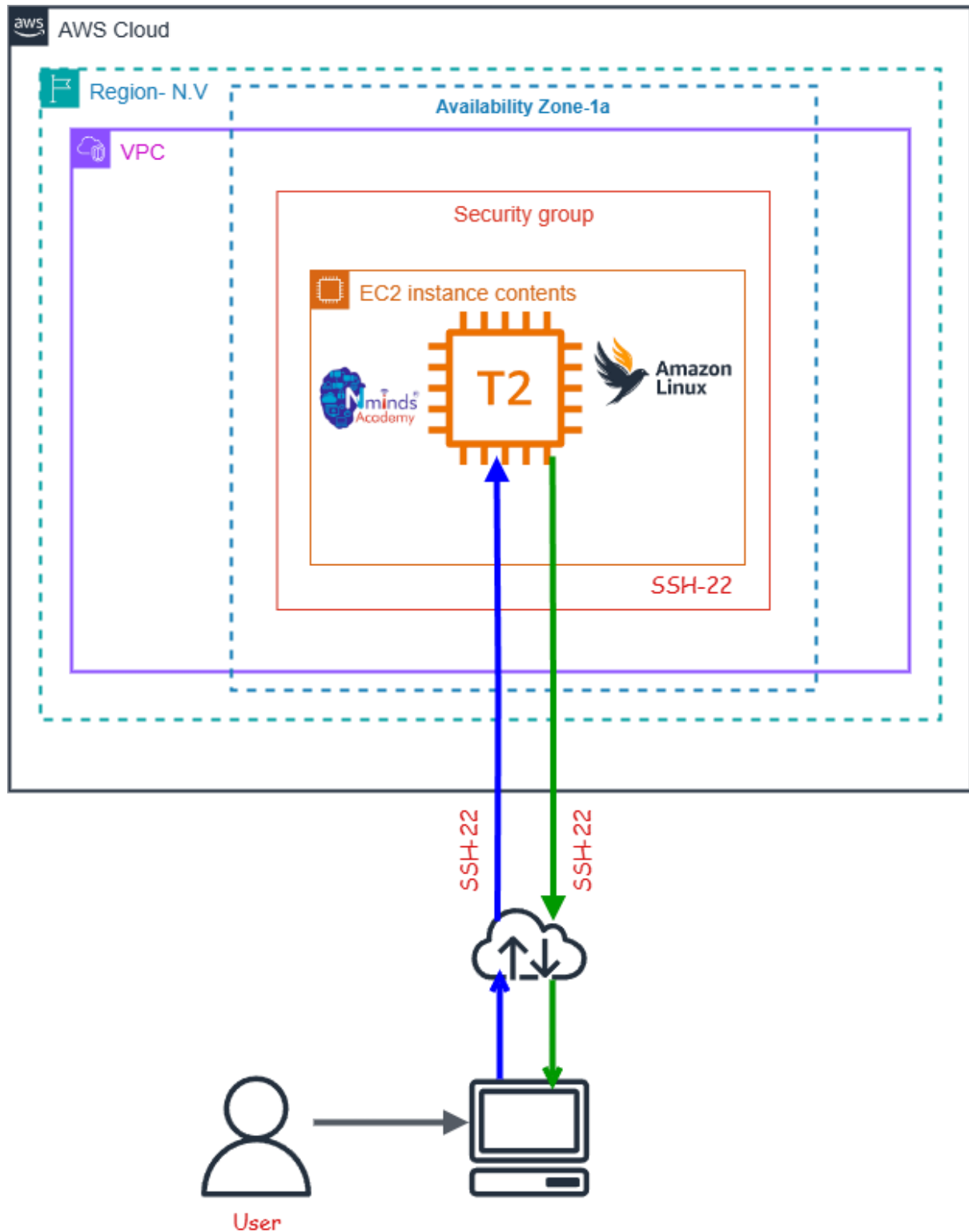
Used Services in AWS



Amazon EC2



Topology



Execution Tasks:



aws

Search

[Alt+S]

Asia Pacific (Sydney)

Amirtha (3604-3528-3604)

Amirtha

EC2 > Instances > Launch an instance

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

[Add additional tags](#)

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-0ac4101c751ee35f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#)[Launch instance](#)[Preview code](#)

aws

Search

[Alt+S]

Asia Pacific (Sydney)

Amirtha (3604-3528-3604)

Amirtha

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-077228dc55d56063d)

Launch log

Next Steps

<

1

2

3

4

5

6

>

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

[Create billing alerts](#)

Connect to your instance

Once your instance is running, log into it from your local computer.

[Connect to instance](#)[Learn more](#)

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

[Connect an RDS database](#)[Create a new RDS database](#)[Learn more](#)

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

[Create EBS snapshot policy](#)

Manage detailed monitoring

Create Load Balancer

Create AWS budget

Manage CloudWatch alarms

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aws

Search

[Alt+S]

Asia Pacific (Sydney)

Amirtha (3604-3528-3604)

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Instances (1/2) info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	Amirtha	i-0cbde35c647f2de47	Running	t3.micro	3/3 checks passed	View alarms +	ap-southeast-2c	ec2-52-62-41-21.ap-southeast-2.compute.amazonaws.com open address
☒	varshini	i-077228dc55d56063d	Running	t3.micro	Initializing	View alarms +	ap-southeast-2c	ec2-3-27-41-21.ap-southeast-2.compute.amazonaws.com open address

i-077228dc55d56063d (varshini)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary info

Instance ID

i-077228dc55d56063d

Public IPv4 address

3.27.41.21 | open address

Private IPv4 addresses

172.31.20.126

Instance state

Running

Public DNS

ec2-3-27-41-21.ap-southeast-2.compute.amazonaws.com | open address

https://ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#Instances...

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AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

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5.Configure Apache Web Server on Linux Instance in AWS



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Objective

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An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache, FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- **Web and Application Hosting:** Run Apache web servers or other Linux-based web applications in the cloud.
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- **Remote Desktop Applications:** Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- **Development and Testing:** Create development environments with Linux OS to test PHP, MySQL or other Linux-based applications.

Used Services in AWS



Amazon EC2

Used Services in Linux Server

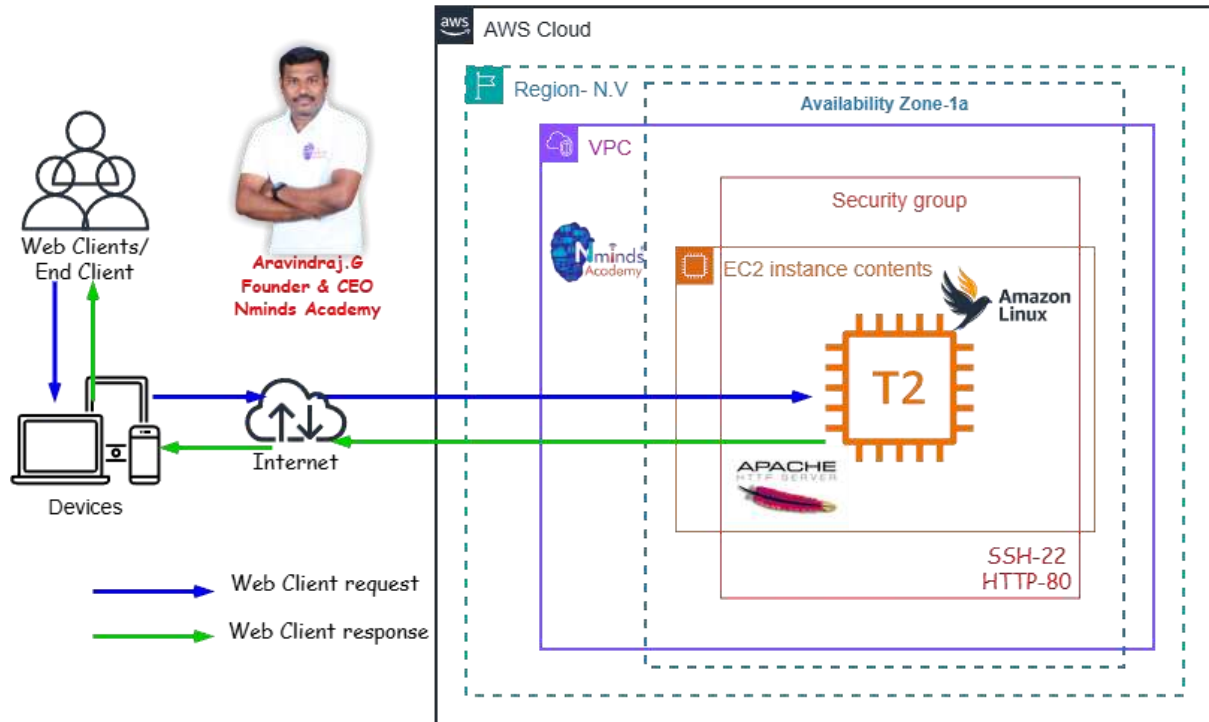


Apache Webserver



Topology

How to Configure the Apache web server in Linux Server with AWS EC2



Execution Tasks:

The screenshot displays the AWS Management Console interface for the Asia Pacific (Sydney) region. The top navigation bar includes the AWS logo, search functionality, and user information (Amirtha). The left-hand sidebar lists various services, with 'EC2' selected under the 'Instances' category.

The main content area shows the 'Instances (1/2)' page. It features a table listing two EC2 instances:

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	Amirtha	i-0cbde35c647f2de47	Running	t3.micro	3/3 checks pass	View alarms +	ap-southeast-2c	ec2-52-62-z...
☒	varshini	i-077228dc55d56063d	Running	t3.micro	Initializing	View alarms +	ap-southeast-2c	ec2-3-27-41...

Below the table, the details for instance 'i-077228dc55d56063d (varshini)' are expanded. The 'Details' tab is active, showing the following information:

- Instance summary Info**
 - Instance ID: i-077228dc55d56063d
 - IPv6 address: -
- Public IPv4 address**: 3.27.41.21 | open address
- Private IPv4 addresses**: 172.31.20.126
- Instance state**: Running
- Public DNS**: ec2-3-27-41-21.ap-southeast-2.compute.amazonaws.com | open address

The bottom of the console shows the URL bar with the path: https://ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#instances.

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\amirt>cd downloads

C:\Users\amirt\Downloads>ssh -i Varsh.pem ec2-user@3.27.41.21
The authenticity of host '3.27.41.21 (3.27.41.21)' can't be established.
ED25519 key fingerprint is SHA256:teNGDB0HjcKzSSxIdbl6+LBnhDLgQcuYso5mX+HOIlg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.27.41.21' (ED25519) to the list of known hosts.
```

```
# Amazon Linux 2023
~ \_ #####
nn \_#####\
nn   \|###|
nn    \|#/ --- https://aws.amazon.com/linux/amazon-linux-2023
      V ~ ' '~>
        |
       / 
      /  
     /   
    /    
   /      
  /        
_/m/'
```

[ec2-user@ip-172-31-20-126 ~]\$ sudo -i
[root@ip-172-31-20-126 ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository 144 kB/s | 30 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-20-126 ~]# yum install httpd -y
Last metadata expiration check: 0:02:26 ago on Tue Mar 10 09:13:01 2026.
Dependencies resolved.



```
root@ip-172-31-20-126/var/n × + v
Nothing to do.
Complete!
[root@ip-172-31-20-126 ~]# yum install httpd -y
Last metadata expiration check: 0:02:26 ago on Tue Mar 10 09:13:01 2026.
Dependencies resolved.
=====
Package                Architecture    Version                               Repository    Size
=====
Installing:
httpd                  x86_64         2.4.66-1.amzn2023.0.1               amazonlinux    47 k
Installing dependencies:
apr                    x86_64         1.7.5-1.amzn2023.0.4               amazonlinux    129 k
apr-util               x86_64         1.6.3-1.amzn2023.0.2               amazonlinux    97 k
apr-util-lmdb          x86_64         1.6.3-1.amzn2023.0.2               amazonlinux    13 k
generic-logos-httpd    noarch         18.0.0-12.amzn2023.0.3             amazonlinux    19 k
httpd-core             x86_64         2.4.66-1.amzn2023.0.1               amazonlinux    1.4 M
httpd-filesystem        noarch         2.4.66-1.amzn2023.0.1               amazonlinux    13 k
httpd-tools            x86_64         2.4.66-1.amzn2023.0.1               amazonlinux    81 k
libbrotli              x86_64         1.0.9-4.amzn2023.0.2               amazonlinux    315 k
mailcap                noarch         2.1.49-3.amzn2023.0.3              amazonlinux    33 k
Installing weak dependencies:
apr-util-openssl       x86_64         1.6.3-1.amzn2023.0.2               amazonlinux    15 k
mod_http2              x86_64         2.0.27-1.amzn2023.0.3              amazonlinux    166 k
mod_lua                x86_64         2.4.66-1.amzn2023.0.1               amazonlinux    60 k
Transaction Summary
=====
Install 13 Packages
Total download size: 2.4 M
```



It works!

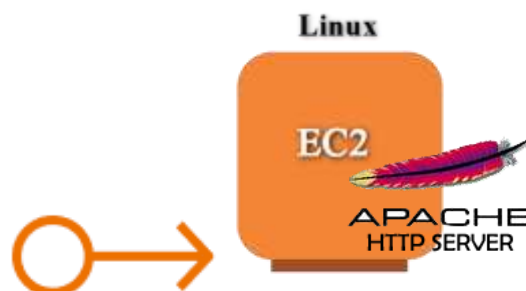




AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

**Trainer: Aravindraj.G- Nminds Academy AWS
Academy Certified Educator**

6.Configure Elastic IP Address to Windows Web Server in AWS



Amirthavarshini S
CSE-Artificial Intelligence & Machine Learning,
Sri Eshwar College of Engineering,
Coimbatore.



Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

1. Highly Available Applications:

- If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.

2. Web Servers:

- If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.

3. Disaster Recovery:

- Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

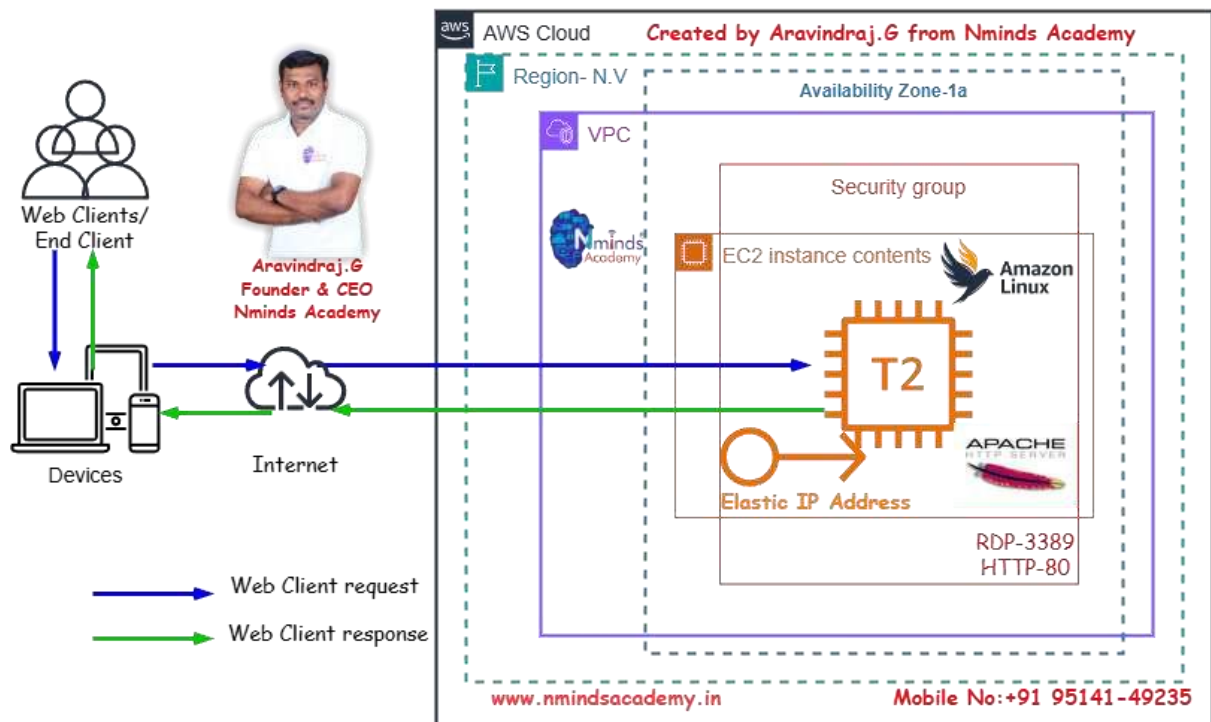
Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.



Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



Used Services in AWS

Amazon EC2

Used

Services in Windows Server

Apache Webserver

APACHE
HTTP SERVER

Amazon Elastic IP Address



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Execution Tasks:

The screenshot shows the AWS console interface for associating an Elastic IP address. The breadcrumb trail is EC2 > Elastic IP addresses > Associate Elastic IP address. The page title is 'Associate Elastic IP address' with an 'Info' link. Below the title, it says 'Choose the instance or network interface to associate to this Elastic IP address (15.135.117.35)'. The 'Elastic IP address: 15.135.117.35' is displayed. Under 'Resource type', 'Instance' is selected. A warning box states: 'If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. Learn more.' It also notes: 'If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.' The 'Instance' field contains 'i-077228dc5d56063d'. The 'Private IP address' field contains '172.31.20.126'. The 'Reassociation' checkbox 'Allow this Elastic IP address to be reassociated' is unchecked.

Associate Elastic IP address [Info](#)

Choose the instance or network interface to associate to this Elastic IP address (15.135.117.35)

Elastic IP address: 15.135.117.35

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

Warning: If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance
i-077228dc5d56063d

Private IP address
The private IP address with which to associate the Elastic IP address.
172.31.20.126

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
☐ Allow this Elastic IP address to be reassociated

The screenshot shows the AWS console interface for allocating an Elastic IP address. The breadcrumb trail is EC2 > Elastic IP addresses > Allocate Elastic IP address. The page title is 'Allocate Elastic IP address' with an 'Info' link. Under 'Elastic IP address settings', 'Public IPv4 address pool' is selected, with 'Amazon's pool of IPv4 addresses' chosen. Other options are disabled. The 'Network border group' is set to 'ap-southeast-2'. A section for 'Global static IP addresses' mentions AWS Global Accelerator. A 'Tags - optional' section states 'No tags associated with the resource' and includes an 'Add new tag' button.

Allocate Elastic IP address [Info](#)

Elastic IP address settings [Info](#)

Public IPv4 address pool

☒ Amazon's pool of IPv4 addresses
☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [Info](#)
ap-southeast-2

Global static IP addresses
AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)
[Create accelerator](#)

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 more tag



Instances (1/2)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
arul	i-0e76b0b628576e90	Running	t3.micro	3/3 checks passed	View alarms +	ap-southeast-2c	ec2-15-13
selvi	i-0e8879340b515c592	Running	t3.micro	3/3 checks passed	View alarms +	ap-southeast-2c	ec2-52-68

i-0e8879340b515c592 (selvi)

Name	Security group rule ID	Port range	Protocol	Source	Security group
-	sgp-0709eb0b1ad216c5	80	TCP	0.0.0.0/0	saure-hy-wdsd-3
-	sgp-056a8dc5e8122628d	22	TCP	0.0.0.0/0	saure-hy-wdsd-3

My Portfolio

Welcome to my personal website

Home About Projects Contact

About Me

Hello! I am a student learning web development. I enjoy coding and building projects.

My Projects

1. Task Tracker Web App
2. Online Book Selling Platform
3. Credit Card Fraud Detection System

Contact

Email: example@email.com





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Academy Certified Educator**

7.Configure SFTP to Linux Web Server in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

- **Web and Application Hosting:** Run Apache web servers or other Linux-based web applications in the cloud.
- **Database Servers:** Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- **Remote Desktop Applications:** Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- **Development and Testing:** Create development environments with Linux OS to test PHP, MySQL or other Linux-based applications. SFTP- Secure File Transfer Protocol

SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

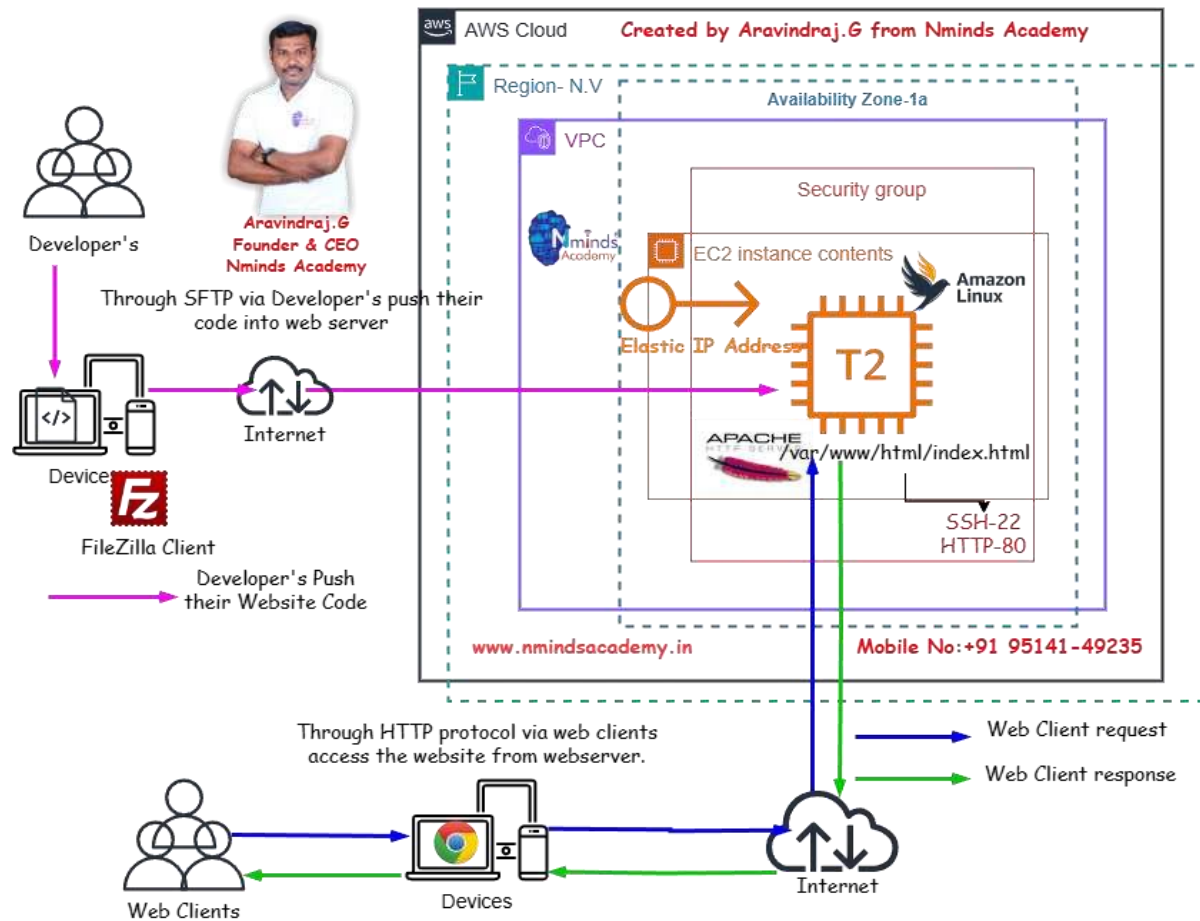
Here are the key uses and advantages of SFTP:

1. **File Transfer:** SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. **File Management:** Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. **Secure Authentication:** SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. **Encryption:** The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.
5. **Firewall-Friendly:** SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.



Topology

How to Configure Developer's push their website code by using SFTP Protocol in Linux Server with AWS EC2 with Elastic IP Address



Used Services in AWS
Services in Windows Server

Used



APACHE
HTTP SERVER

Apache Webserver



```
Microsoft Windows [Version 10.0.26200.7840]
(C) Microsoft Corporation. All rights reserved.

C:\Users\amirt>ssh -i Varsh.pem ec2-user@15.135.117.35
Warning: Identity file Varsh.pem not accessible: No such file or directory.
ec2-user@15.135.117.35: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

C:\Users\amirt>cd downloads

C:\Users\amirt\Downloads>ssh -i Varsh.pem ec2-user@15.135.117.35
```

The terminal shows the user navigating through various prompts during the SSH login process:

```
#_#####
#_#####|
#\#####}
#/#/\---> https://aws.amazon.com/linux/amazon-linux-2023
#/V##!
#/_/
#/_/!
```

```
Last login: Tue Mar 10 09:37:45 2026 from 121.200.55.210
[ec2-user@ip-172-31-20-126 ~]$ useradd amirtha
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
[ec2-user@ip-172-31-20-126 ~]$ sudo -i
[root@ip-172-31-20-126 ~]# useradd amirtha
[root@ip-172-31-20-126 ~]# useradd varshini
[root@ip-172-31-20-126 ~]# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534::Kernel Overflow User::-/sbin/nologin

rpcuser:x:29:
tcpdump:x:72:
screen:x:84:
ec2-user:x:1000:
apache:x:48:
amirtha:x:1001:
varshini:x:1002:
mathi:x:1003:amirtha,varshini
[root@ip-172-31-20-126 ~]# passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-20-126 ~]# passwd ec2-user
Changing password for user ec2-user.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-20-126 ~]# passwd amirtha
Changing password for user amirtha.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-20-126 ~]# passwd varshini
Changing password for user varshini.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-20-126 ~]# ls -ld /var/www/html
drwxr-xr-x. 2 root root 24 Mar 10 09:22 /var/www/html
[root@ip-172-31-20-126 ~]# chgrp mathi /var/www/html
[root@ip-172-31-20-126 ~]# ls -ld /var/www/html
drwxr-xr-x. 2 root mathi 24 Mar 10 09:22 /var/www/html
[root@ip-172-31-20-126 ~]# chmod 775 /var/www/html
[root@ip-172-31-20-126 ~]# ls -ld /var/www/html
drwxrwxr-x. 2 root mathi 24 Mar 10 09:22 /var/www/html
[root@ip-172-31-20-126 ~]# vi /etc/ssh/sshd_config
[root@ip-172-31-20-126 ~]# systemctl reload sshd
[root@ip-172-31-20-126 ~]# |
```



Amazon Elastic IP Address



sftp://amirtha@15.135.117.35 - FileZilla

File Edit View Transfer Server Bookmarks Help

Host: sftp://15.135.117.35 Username: amirtha Password: Port: Quickconnect

Status: Connected to 15.135.117.35
 Status: Retrieving directory listing...
 Status: Listing directory /home/amirtha
 Status: Directory listing of "/home/amirtha" successful

Local site: C:\Users\amirt\ Remote site: /home/amirtha

Filename	Filesize	Filetype	Last modified
..			
.android		File folder	07-12-2025 15:...
.antigravity		File folder	04-03-2026 14:...
.arduinoIDE		File folder	19-12-2025 09:...
.aws		File folder	18-12-2025 12:...
.cache		File folder	13-09-2025 06:...
.codeium		File folder	07-03-2026 13:...
.conda		File folder	14-08-2025 06:...
.config		File folder	10-12-2025 12:...
.copilot		File folder	26-02-2026 13:...
.eclipse		File folder	02-04-2024 22:...

14 files and 68 directories. Total size: 30,370,496 bytes

Filename	Filesize	Filetype	Last modified	Permissi...	Owner/Gr...
..					
.bash_logout	18	BASH_L...	29-01-2023...	-rw-r--r--	amirtha a...
.bash_profile	141	BASH_P...	29-01-2023...	-rw-r--r--	amirtha a...
.bashrc	492	BASHRC...	29-01-2023...	-rw-r--r--	amirtha a...

3 files. Total size: 651 bytes

Server/Local file Direc... Remote file Size Priority Status

Queued files Failed transfers Successful transfers

Queue: empty

sftp://amirtha@15.135.117.35 - FileZilla

File Edit View Transfer Server Bookmarks Help New version available!

Host: sftp://15.135.117.35 Username: amirtha Password: Port: Quickconnect

Status: File transfer successful, transferred 1,933 bytes in 1 second
 Status: Retrieving directory listing of "/var/www/html"...
 Status: Listing directory /var/www/html
 Status: Directory listing of "/var/www/html" successful

Local site: C:\Users\amirt\Downloads\ Remote site: /var/www/html

Filename	Filesize	Filetype	Last modified
Heart Monitoring A...	201,761	Compressed (z...	05-11-2025 04:...
index.html	1,246	Microsoft Edg...	04-03-2026 15:...
index1.html	1,933	Microsoft Edg...	10-03-2026 13:...
jdk-24_windows-x6...	215,854	Application	02-04-2024 20:...
kidney-history.csv	651	Microsoft Exce...	28-12-2025 08:...
kidney_health_histor...	1,048	Microsoft Exce...	28-12-2025 13:...
kidney_test_2025-12...	220	JSON Source Fi...	28-12-2025 12:...
kiro-ide-0.8.0-stabl...	199,556	Application	28-12-2025 11:...
mingw-get-setup.exe	86,528	Application	26-12-2024 11:...
mongodb-windows...	793,661	Windows Insta...	09-12-2025 18:...
My resume.docx	33,213	Microsoft Wor...	13-07-2025 20:...

Selected 1 file. Total size: 1,933 bytes

Filename	Filesize	Filetype	Last modified	Permissi...	Owner/Gr...
..					
index.html	1,838	Microsof...	10-03-2026...	-rw-r--r--	root root
index1.html	1,933	Microsof...	10-03-2026...	-rw-rw-r--	amirtha a...

2 files. Total size: 3,771 bytes

Server/Local file Direc... Remote file Size Priority Status



